

Power Cell Escape — Concept & Design (First Version)

1. Theme and feel

I'm making a small 2D game where you play a little robot stuck in an old power plant. To escape, you have to find three power cells and bring them to the exit.

I want the game to feel a bit tense but fair. You have 180 seconds. You collect cells while avoiding one enemy that moves around. The screen always shows how many cells you've got, your lives, and the time left.

When something happens — you grab a cell, get hit, unlock the exit, win or lose — the game gives you a sound and a flash so you know.

I also made sure the game isn't colour-only: the cells are labelled B1, B2, B3, the exit says LOCKED or OPEN, and the enemy has a "!" above it.

2. How the game plays

- You start, read the instructions, then the game begins.
- Walk over a cell to collect it. The count goes up.
- When you have all three, the exit unlocks.
- If the enemy touches you, you lose a life. You blink for 1.5 seconds so it can't hit you again right away.
- If the time runs out, or you lose all three lives, you lose.
- If you reach the exit after it's open, you win.
- Press R at any time to start over.

3. Controls and rules

Controls

Key	Action
WASD or arrow keys	move
R	restart
C	high-contrast mode
M	mute
[and]	turn volume down / up
I	show or hide help

Rules

The room has walls and a few blocks inside. Three cells are placed around it. The exit stays locked (red) until all three cells are collected, then it turns green and opens. The enemy is a red square that walks a set path; if you get close, it chases you for a short while, then goes back to its path. You start with three lives.

4. Winning and losing

- Win: collect all three cells and reach the open exit.
- Lose: run out of lives, or run out of time.
- Restart anytime with R.

5. The level

The room is about 21 by 15 units. You start in the bottom-left. The exit is in the bottom-right. The three cells are spread out — one top-left, one top-right, one near the middle. The enemy walks a loop in the upper-middle of the room, so you have to time your moves. A few wall blocks make you take longer routes and give you places to hide.

6. How I'm building it

I'm using Unity 2022.3 (2D). Most of the game is built in code when it starts, so I don't rely on hand-made scene files. The main scripts:

- GameManager — keeps the game state, counts cells and lives, runs the timer, decides win/lose.
- PlayerController — movement and taking damage.
- Battery and ExitDoor — the cells and the exit.
- PatrollingEnemy — the enemy's movement.
- LevelBuilder and GameBootstrap — put the whole level together at start.
- HUDManager and InstructionsPanel — the text and info on screen.
- AudioFeedback, SettingsManager, GameArt — sounds, settings, and the simple shapes.

7. My plan (7 days)

Day	What I'll do
Day 1 (7/29)	set up the project, write this doc, start the main scripts
Day 2 (7/30)	enemy, sound, settings, build the level
Day 3 (7/31)	HUD, help screen, fix bugs in the loop
Day 4 (8/1)	accessibility and balance (make it fair)
Day 5 (8/2)	testing — collection, unlock, win/lose, restart — and fill the test table

Day 6 (8/3)	ask a friend to play, record the video, write the report
Day 7 (8/4–8/5)	final build, tag the release, check everything

I'll update the dev log every day, not just at the end.

8. What I'm using

- Unity 2022.3 LTS (2D).
- All art and sound are made in code: simple shapes for graphics and short beeps for sound. No outside files.
- No extra libraries.
- If I add anything made by someone else, I'll say so and credit it.

9. Legal, ethics, accessibility, safety

- Legal: I'm not using copied art, audio, or code. Anything I learn from a tutorial will be credited. If AI helps me write or understand code, I'll say that in the report.
- Accessibility: a high-contrast mode (C), volume control and mute (M, []), text labels instead of colour alone, and large readable text.
- Safety: it's a single-player offline game. 10. Risks and what I'll do

Risk	What I'll do
Project might not open (wrong Unity version)	state the version clearly and include the project settings
Exit unlocks too early by mistake	test the order of collecting and unlocking
Game too hard or too easy	playtest and adjust
Running low on time	follow the plan and finish a working loop before polishing